

Chapter 1

General Concepts

1. Dual Hypergraphs

Let $X = \{x_1, x_2, \dots, x_n\}$ be a finite set. A *hypergraph* on X is a family $H = (E_1, E_2, \dots, E_m)$ of subsets of X such that

$$(1) \quad E_i \neq \emptyset \quad (i = 1, 2, \dots, m)$$

$$(2) \quad \bigcup_{i=1}^m E_i = X.$$

A *simple hypergraph* (or "Sperner family") is a hypergraph $H = (E_1, E_2, \dots, E_m)$ such that

$$(3) \quad E_i \subset E_j \Rightarrow i = j$$

The elements $x_1, x_2, \dots, x_n$ of X are called *vertices*, and the sets $E_1, E_2, \dots, E_m$ are the *edges* of the hypergraph. A simple graph is a simple hypergraph each of whose edges has cardinality 2; a *multigraph* (with loops and multiple edges) is a hypergraph in which each edge has cardinality ≤ 2 . Nonetheless we shall not consider isolated points of a graph to be vertices.

A hypergraph H may be drawn as a set of points representing the vertices. The edge E_j is represented by a continuous curve joining the two elements if $|E_j| = 2$, by a loop if $|E_j| = 1$, and by a simple closed curve enclosing the elements if $|E_j| \geq 3$.

One may also define a hypergraph by its incidence matrix $A = ((a_j^i))$, with columns representing the edges $E_1, E_2, \dots, E_m$ and rows representing the vertices $x_1, x_2, \dots, x_n$, where $a_j^i = 0$ if $x_i \notin E_j$, $a_j^i = 1$ if $x_i \in E_j$ (cf. Figure 1).

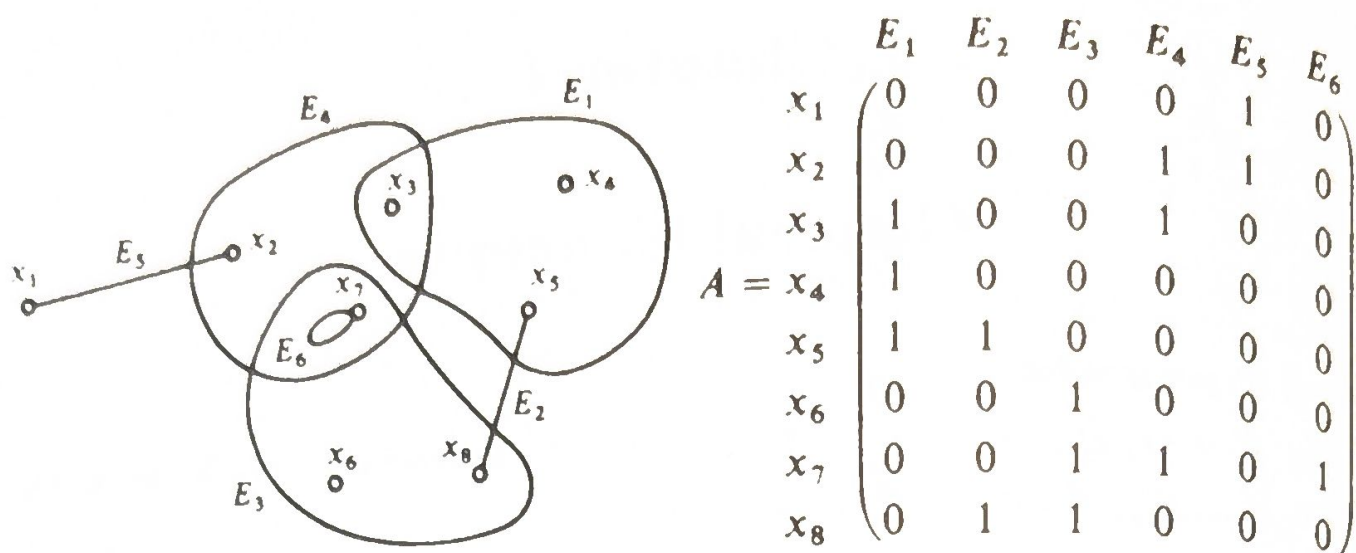


Figure 1. Representation of a hypergraph H and its incidence matrix

The dual of a hypergraph $H = (E_1, E_2, \dots, E_m)$ on X is a hypergraph $H^* = (X_1, X_2, \dots, X_n)$ whose vertices $e_1, e_2, \dots, e_m$ correspond to the edges of H , and with edges

$$X_i = \{e_j / x_i \in E_j \text{ in } H\}$$

H^* clearly satisfies both conditions (1) and (2).

It is easily seen that the incidence matrix of H^* is the transpose of the incidence matrix of H and so we have $(H^*)^* = H$.

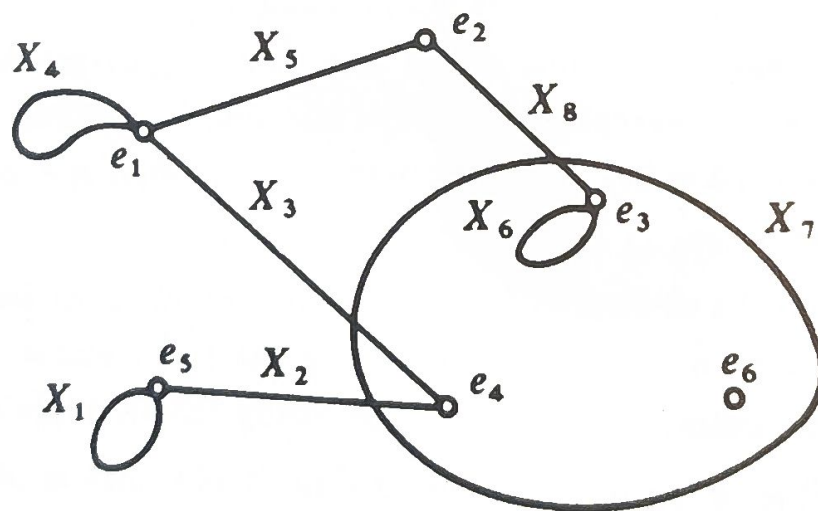


Figure 2. The dual hypergraph of the hypergraph in Figure 1.

As for a graph, the order of H , denoted by $n(H)$, is the number of vertices. The number of edges will be denoted by $m(H)$. Further the rank is $r(H) = \max_j |E_j|$, the anti-rank is $s(H) = \min_j |E_j|$; if $r(H) = s(H)$ we say that H is a uniform hypergraph;

a simple uniform hypergraph of rank r will also be called r -uniform, and in this case it will be understood that there is no repeated edge.

For a set $J \subset \{1, 2, \dots, m\}$ we call the family

$$H' = (E_j / j \in J)$$

the *partial hypergraph generated by the set J* . The set of vertices of H' is a nonempty subset of X .

For a set $A \subset X$ we call the family

$$H_A = (E_j \cap A / 1 \leq j \leq m, E_j \cap A \neq \emptyset)$$

the *sub-hypergraph induced by the set A* . (We define *partial sub-hypergraphs* etc. in a similar fashion).

Proposition. *The dual of a subhypergraph of H is a partial hypergraph of the dual hypergraph H^* .*

In the case of hypergraphs of rank 2 these reduce to the familiar definitions for graphs. All the concepts of graph theory may thus be generalised to hypergraphs which will allow us to find stronger theorems, and applications to objects other than graphs. Further the formulation of a combinatorial problem in terms of hypergraphs sometimes has the advantage of providing a remarkably simple statement having a familiar form.

A stronger result may be much easier to prove than the weak result!

2. Degrees

The other definitions from graph theory which may be extended without ambiguity to a hypergraph H are the following:

For $x \in X$, define the *star* $H(x)$ with centre x to be the partial hypergraph formed by the edges containing x . Define the degree $d_H(x)$ of x to be the number of edges of $H(x)$, so $d_H(x) = m(H(x))$.

The maximum degree of the hypergraph H will always be denoted by

$$\Delta(H) = \max_{x \in X} d_H(x).$$

A hypergraph in which all vertices have the same degree is said to be *regular*.

Note that $\Delta(H) = r(H^*)$, and that the dual of a regular hypergraph is uniform.

For a hypergraph H of order n , the degrees $d_H(x_i) = d_i$ in decreasing order form an n -tuple $d_1 \geq d_2 \geq \dots \geq d_n$ whose properties can be characterised if H is a simple graph (Erdős, Gallai [1960], cf. *Graphs*, Ch. 6, Th. 6). In general

Proposition 1. *An n -tuple $d_1 \geq d_2 \geq \dots \geq d_n$ is the degree sequence of a uniform hypergraph of rank r and order n (possibly with repeated edges) if and only if $\sum_{i=1}^n d_i$ is a multiple of r and $d_n \geq 1$.*

Proof. Given such an n -tuple $d_1 \geq d_2 \geq \dots \geq d_n$, we wish to construct the edges of a hypergraph H one by one on the set $\{x_1, x_2, \dots, x_n\}$.

In the first step, associate with each vertex x_i a weight $d_i^1 = d_i$ and form the first edge E_1 by taking the r vertices of greatest weight. In the second step, associate with vertex x_i the weight

$$d_i^2 = \begin{cases} d_i^1 & \text{if } x_i \notin E_1 \\ d_i^1 - 1 & \text{if } x_i \in E_1 \end{cases}$$

Form E_2 by taking the r vertices of greatest weight, etc. If $\sum d_i = mr$ we obtain H with the edges $E_1, E_2, \dots, E_m$, and $d_H(x_i) = d_i$ for $i = 1, 2, \dots, n$.

A hypergraph is *connected* if the intersection graph of the edges is connected. Then we have

Proposition 2 (Tuszyadej [1978]). *An n -tuple $d_1 \geq d_2 \geq \dots \geq d_n$ is the degree sequence of a connected uniform hypergraph of rank r if and only if*

$$(1) \quad \sum_{i=1}^n d_i \text{ is a multiple of } r,$$

$$(2) \quad d_i \geq 1 \quad (i = 1, 2, \dots, n),$$

$$(3) \quad \sum_{i=1}^n d_i \geq \frac{r(n-1)}{r-1},$$

$$(4) \quad d_1 \leq m = \frac{\sum d_i}{r}.$$

(For extensions to non-uniform hypergraphs, cf. Boonyasombat [1984]).

Theorem 1 (Gale [1957], Ryser [1957]). *Given m integers $r_1, r_2, \dots, r_m$ and an n -tuple of integers $d_1 \geq d_2 \geq \dots \geq d_n$, there exists a hypergraph $H = (E_1, E_2, \dots, E_m)$ on a set $X = \{x_1, x_2, \dots, x_n\}$ such that $d_H(x_i) = d_i$ for $i \leq n$ and $|E_j| = r_j$ for $j \leq m$ if and only if*

$$(1) \quad \sum_{j=1}^m \min\{r_j, k\} \geq d_1 + d_2 + \dots + d_k \quad (k < n)$$

$$(2) \quad \sum_{j=1}^m r_j = d_1 + d_2 + \dots + d_n.$$

Proof. We deduce this immediately from the theory of network flows (corollary to theorem 3, Ch.5 in *Graphs*). Indeed, construct a network flow with vertices the points $j = 1, 2, \dots, m$ and $x_1, x_2, \dots, x_n$, with a source a and a sink z . The arcs are

- all arcs (a, j) with capacity r_j
- all arcs (x_i, z) with capacity d_i
- all arcs (j, x_i) with capacity 1.

It suffices to show that there exists an integer flow satisfying the capacities, saturating each of the arcs (j, z) entering the sink z , that is to say that the maximum flow which can enter set $\{x_i / i \in I\}$ is always greater than or equal to the sum $\sum_{i \in I} d_i$, for

all $I \subset \{1, 2, \dots, n\}$. (Further, we note that thanks to the network flow theorem we may always suppose that such a flow never leaves empty an "entry" arc or an "exit" arc.)

Open Problem. *Find a necessary and sufficient condition for an m -tuple (r_j) and an n -tuple (d_i) to be respectively the $|E_j|$ and the $d_H(x_i)$ of a simple hypergraph H .*

Let r, n be integers, $1 \leq r \leq n$. We define the r -uniform complete hypergraph of order n (or the r -complete hypergraph) to be a hypergraph denoted K_n^r consisting of all the r -subsets of a set X of cardinality n . We may now state in a complete form the celebrated *Sperner's theorem* [1926]; in fact the inequality (1), which allows for a

simple proof was discovered (independently) much later by Yamamoto, Meshalkin, Lubell and Bollobás.

Theorem 2 (Sperner [1928]; proof by Yamamoto, Meshalkin, Lubell, Bollobás). *Every simple hypergraph H of order n satisfies*

$$(1) \quad \sum_{E \in H} \binom{n}{|E|}^{-1} \leq 1.$$

Further, the number of edges $m(H)$ satisfies

$$(2) \quad m(H) \leq \binom{n}{\lfloor n/2 \rfloor}$$

For $n = 2h$ even, equality in (2) is attained if and only if H is the hypergraph K_n^h . For $n = 2h - 1$ odd, equality in (2) is attained if and only if H is the hypergraph K_n^h or the hypergraph K_n^{h+1} .

Proof. Let X be a finite set of cardinality n . Consider a directed graph G with vertices the subsets of X , and with an arc from $A \subset X$ to $B \subset X$ if $A \subset B$ and $|A| = |B| - 1$.

Let $E \in H$, the number of paths in the graph G from the vertex $\emptyset$ to the vertex E is $|E|!$, thus the total number of paths from $\emptyset$ to X is $n! \geq \sum_{E \in H} (|E|!)(n - |E|!)$ (as

H is a simple hypergraph, a path passing through E cannot pass through $E' \in H$, $E' \neq E$). We thus deduce inequality (1).

For the second part,

$$\binom{n}{|E|} \leq \binom{n}{\lfloor n/2 \rfloor}.$$

whence

$$1 \geq \sum_{E \in H} \binom{n}{|E|}^{-1} \geq m(H) \binom{n}{\lfloor n/2 \rfloor}^{-1}.$$

We immediately deduce inequality (2).

Let H be a hypergraph satisfying equality in (2). Then for all $E \in H$,

$$(3) \quad \binom{n}{|E|} = \binom{n}{\lfloor n/2 \rfloor}.$$

If $n = 2h$ is even, (3) implies that H is h -uniform, and since $m(H) = \binom{n}{h}$ we have $H = K_n^h$, and the proof is achieved.

If $n = 2h+1$, (3) implies that $h \leq |E| \leq h+1$ for all $E \in H$. Let X_k be the set of vertices in G which represent edges of H with cardinality k ; the set $X_h \cup X_{h+1}$ is a stable set of G , and $m(H) = |X_h \cup X_{h+1}|$.

The number of arcs of G leaving X_h is equal to $|X_h|(n-h)$; the number of arcs entering the image ΓX_h of X_h is $|\Gamma X_h|(h+1)$. Thus

$$|\Gamma X_h|(h+1) \geq |X_h|(n-h),$$

or

$$|\Gamma X_h| \geq \frac{2h+1-h}{h+1} |X_h| = |X_h|.$$

If X_h is non-empty and is not the set $\mathcal{P}_h(X)$ of all h -subsets of X , the above inequality is strict (because the bipartite subgraph of G generated by the h -subsets and $(h+1)$ -subsets is connected), whence

$$\begin{aligned} m(H) &= |X_h| + |X_{h+1}| \leq |X_h| + |\mathcal{P}_{h+1}(X) - \Gamma X_h| \\ &< |X_h| + \binom{n}{h+1} - |X_h| = \binom{n}{h+1}. \end{aligned}$$

Thus, equality in (2) is possible only if $X_h = \emptyset$ or $X_h = \mathcal{P}_h(X)$, i.e. if $H = K_n^h$ or K_n^{h+1} .

Q.E.D.

For extensions of Theorem 2 see: Erdős [1945], Kleitman [1968], Meshalkin [1963], Kleitman [1965], Greene, Kleitman [1976], Katona [1966], Hochberg, Hirsch [1970], Erdős, Frankl, Katona [1984].

To generalise graphs without "pendent" vertices, we consider the following class of hypergraphs; a hypergraph H is said to be *separable* if for every vertex x , the intersection of the edges containing x is the singleton $\{x\}$ i.e. if $\bigcap_{E \in H(x)} E = \{x\}$.

Corollary. *If an n -tuple $d_1 \geq d_2 \geq \dots \geq d_n$ of positive integers is the degree sequence of a separable hypergraph $H = (E_1, \dots, E_m)$ then*

$$\sum_{i=1}^n \binom{m}{d_i}^{-1} \leq 1.$$

Essentially H is separable if and only if its dual H^* is a simple hypergraph, which implies, by Theorem 2,

$$\sum_{i=1}^n \binom{m}{|X_i|}^{-1} \leq 1.$$

Q.E.D.

To generalise simple graphs, we say that a hypergraph $H = (E_1, E_2, \dots, E_m)$ is *linear* if $|E_i \cap E_j| \leq 1$ for $i \neq j$. For example, the hypergraphs of Figures 1,2 are linear.

We have immediately

Proposition 3. *The dual of a linear hypergraph is also linear.*

Indeed, if H is linear, two edges X_i and X_j in H^* cannot intersect in two distinct points e_1, e_2 , as then, in H , $E_1 \supset \{x_1, x_2\}$, $E_2 \supset \{x_1, x_2\}$, contradicting $|E_1 \cap E_2| \leq 1$.

Theorem 3. *For every linear hypergraph H of order n , we have*

$$(1) \quad \sum_{E \in H} \binom{|E|}{2} \leq \binom{n}{2}.$$

If in addition, H is r -uniform, then the number of edges satisfies

$$(2) \quad m(H) \leq \frac{n(n-1)}{r(r-1)}.$$

The bound in (2) is attained if and only if H is a Steiner system $S(2, r, n)$.

For, the number of pairs x, y which are contained in a same edge of H is

$$\sum_{E \in H} \binom{|E|}{2} \leq \binom{n}{2}$$

whence we have (1). If H is r -uniform, (2) follows.

A Steiner system $S(2, r, n)$ is an r -uniform hypergraph on X , with $|X| = n$, in which every pair of vertices is contained in exactly one edge. A necessary and sufficient condition for the existence of an $S(2, 3, n)$ system, due to T.P. Kirkman [1847], is

that $n \equiv 1$ or $3 \pmod{6}$.

To exclude some values of r it is easily seen that the following are necessary conditions for the existence of $S(2, r, n)$ systems:

- (1) $\binom{n}{2} \binom{r}{2}^{-1}$ is an integer;
- (2) $(n-1)(r-1)^{-1}$ is an integer.

These conditions are necessary and sufficient for $r = 3, 4$ (Hanani). For $r = 6$ these conditions are sufficient with a single exception: no $S(2, 6, 21)$ system exists. Wilson [1972] has further shown that if r is a prime power and if n is sufficiently large then (1) and (2) are necessary and sufficient.

For all questions on existence and enumeration of $S(2, r, n)$ systems, see Lindner and Rosa [1980]. We give here a list of $S(2, r, n)$ systems known for small values of r and of n :

$S(2, 3, 7)$	
$S(2, 3, 9)$	De Pasquale [1899], Brunel [1901], Cole [1913]
$S(2, 4, 13)$	De Pasquale [1899], Brunel [1901], Cole [1913]
$S(2, 3, 15)$	Cole [1917], White [1919], Fischer [1940]
$S(2, 4, 16)$	Witt [1938]
$S(2, 3, 19)$	Deherder [1976]
$S(2, 3, 21)$	Wilson [1974]
$S(2, 5, 21)$	Witt [1938]
$S(2, 3, 25)$	Wilson [1974]
$S(2, 4, 25)$	Brouwer, Rokowska [1977]
$S(2, 5, 25)$	McInnes [1977]
$S(2, 3, 27)$	McInnes [1977]
$S(2, 4, 28)$	Rokowska [1977]

We deduce that the bound in (2) of Theorem 3 is the best possible for $n = 7$, $r = 3$; or for $n = 9$, $r = 3$; etc.

3. Intersecting Families

Given a hypergraph H , we define an *intersecting family* to be a set of edges having non-empty pairwise intersection. For example, for every vertex x of H , the star $H(x) = \{E/E \in H, x \in E\}$ is an intersecting family of H . The maximum cardinality of an intersecting family, which we denote $\Delta_0(H)$, thus satisfies

$$\Delta_0(H) \geq \max_{x \in X} |H(x)| = \Delta(H).$$

In a multigraph, the intersecting families are just the stars and the triangles (perhaps with multiple edges).

Theorem 4. *Every hypergraph H of order n with no repeated edge satisfies*

$$\Delta_0(H) \leq 2^{n-1}.$$

Further, every maximal intersecting family of the hypergraph of subsets of an n -set has cardinality 2^{n-1} .

Proof. Let $\mathcal{A}$ be a maximal intersecting family of the hypergraph of subsets of X , where $|X| = n$.

If $B_1 \notin \mathcal{A}$ then there exists in $\mathcal{A}$ a set A_1 disjoint from B_1 (by the maximality of $\mathcal{A}$); thus $X - B_1 \supset A_1$, whence, for every $A \in \mathcal{A}$, $(X - B_1) \cap A \neq \emptyset$. By virtue of the maximality of $\mathcal{A}$, we deduce that $(X - B_1) \in \mathcal{A}$. Conversely, if $(X - B_1) \in \mathcal{A}$, we have $B_1 \notin \mathcal{A}$. Hence $B \rightarrow X - B$ is a bijection between $\mathcal{P}(X) - \mathcal{A}$ and $\mathcal{A}$, whence

$$|\mathcal{A}| = \frac{1}{2} |\mathcal{P}(X)| = 2^{n-1}.$$

Lemma (Greene, Katona, Kleitman [1975], anticipated by Bollobás). *Let $x_1, x_2, \dots, x_n$ be points in that order on a circle and let $\mathcal{A} = (A_1, A_2, \dots, A_m)$ be a family of circular intervals of points such that*

$$(1) \quad |A_i| \leq \frac{n}{2} \text{ for all } i \leq m;$$

$$(2) \quad A_i \cap A_j \neq \emptyset \text{ for all } i, j, \quad i \neq j;$$

$$(3) \quad A_i \not\subset A_j \text{ for all } i, j, \quad i \neq j.$$

Then

$$(4) \quad m \leq \min_i |A_i|;$$

$$(5) \quad \sum_{i=1}^m |A_i|^{-1} \leq 1.$$

Equality is attained in (5) if and only if $\mathcal{A}$ is a family of circular intervals of cardinality m each having a point in common.

(*) **Proof.** Let A_1 be a set of minimum cardinality in $\mathcal{A}$. From (2), $A_1 \cap A_i \neq \emptyset$ for $i \neq 1$; and from (3), these $A_1 \cap A_i$ are intervals with one and only one of their ends coinciding with an end of A_1 . From (3) these intervals $A_1 \cap A_i$ are all different. Thus the number of possible intervals of this form is $\leq 2(|A_1| - 1)$. From (1) and (2) two sets $A_1 \cap A_i$ and $A_1 \cap A_j$ with $i \neq j$, $i \neq 1$, $j \neq 1$ cannot constitute a partition of A_1 ; thus only half of these possible intervals can occur, which gives us $m - 1 \leq |A_1| - 1$. Thus, for all i , $|A_i| \geq |A_1| \geq m$, so we have (4) and (5).

Finally, equality in (5) implies

$$1 = \sum_{i=1}^m \frac{1}{|A_i|} \leq \frac{m}{|A_1|} \leq 1$$

So we have $|A_i| = |A_1| = m$, for $1 \leq i \leq m$. Thus the A_i are intervals of length m whose initial end-points are m successive points on the circle. Conversely if the A_i satisfy (1), (2), (3) and are all intervals of length m , then clearly we have equality in (5).

Theorem 5 (Erdős, Chao-Ko, Rado [1961], proof by Greene, Katona, Kleitman [1976]).
Let H be a simple intersecting hypergraph of order n and of rank $r \leq n/2$; then

$$(1) \quad \sum_{E \in H} \binom{n-1}{|E|-1}^{-1} \leq 1;$$

$$(2) \quad m(H) \leq \binom{n-1}{r-1}.$$

Further we have equality in (2) when H is a star of K_n^r (and only then if $r < \frac{n}{2}$).

Proof. Let $X = \{x_1, x_2, \dots, x_n\}$ be the vertex set of H . For any permutation π of $1, 2, \dots, n$, denote by H_π the set of edges of H which are intervals for the circular sequence $x_{\pi(1)}, x_{\pi(2)}, \dots, x_{\pi(n)}, x_{\pi(1)}$. For $E \in H$, put

$$\beta(E) = |\{\pi / E \in H_\pi\}|.$$

From the lemma,

$$(3) \quad \sum_{E \in H_\pi} \frac{1}{|E|} \leq 1.$$

We then have

$$(4) \quad \sum_{E \in H} \frac{\beta(E)}{|E|} = \sum_{E \in H} \sum_{\pi / E \in H_\pi} \frac{1}{|E|} = \sum_{\pi} \sum_{E \in H_\pi} \frac{1}{|E|} \leq n!$$

Let E_0 be an edge of H , with cardinality $|E_0| = h$, and let x_0 be an element of E_0 . Since E_0 is also an edge of the hypergraph $K_n^h(x_0) = H'$, and since from the lemma we have equality in (3) for H' , we have equality in (4) for H' , and

$$\frac{\beta(E_0)}{|E_0|} = \frac{1}{m(H')} \sum_{E' \in H'} \frac{\beta(E')}{|E'|} = \frac{n!}{m(H')} = n! \binom{n-1}{|E_0|-1}^{-1}$$

We may thus write, using (4),

$$\sum_{E \in H} \binom{n-1}{|E|-1}^{-1} = \frac{1}{n!} \sum_{E \in H} \frac{\beta(E)}{|E|} \leq \frac{n!}{n!} = 1.$$

Thus we have (1).

Finally, every $E \in H$ satisfies $|E| \leq r \leq \frac{n}{2}$, so

$$m(H) \binom{n-1}{r-1}^{-1} \leq \sum_{E \in H} \binom{n-1}{|E|-1}^{-1} \leq 1,$$

(2) follows.

Q.E.D.

For extensions to Theorem 5 see Schönheim [1968], Hilton and Milner [1967], Hilton [1979], Erdős, Chao-Ko, Rado [1961], Bollobás [1974], Frankl [1975], Frankl [1976].

If no restriction is made on the rank, then by analogous methods we obtain:

Generalisation (Greene, Kleitman, Katona [1976]). *Let H be a simple hypergraph of order n . If H is intersecting, then*

$$(1) \quad \sum_{\substack{E \in H \\ |E| \leq \frac{n}{2}}} \binom{n}{|E|-1}^{-1} + \sum_{\substack{E \in H \\ |E| > \frac{n}{2}}} \binom{n}{|E|}^{-1} \leq 1.$$

$$(2) \quad m(H) \leq \binom{n}{\lfloor \frac{n}{2} \rfloor + 1}.$$

Further, equality is attained in (2) for $H = K_n^{\lfloor \frac{n}{2} \rfloor + 1}$.

Remark. Theorem 5 shows that

$$\Delta_0(K_n^r) = \begin{cases} \binom{n-1}{r-1} & \text{if } r \leq \frac{n}{2} \\ \binom{n}{r} & \text{if } r > \frac{n}{2} \end{cases}$$

More precisely, we shall show that in the r -complete hypergraph K_n^r , the maximum intersecting families are: for $r < \frac{n}{2}$, the stars of the form $K_n^r(x)$; for $r = \frac{n}{2}$, the maximal intersecting families; for $r > \frac{n}{2}$, the set of edges of K_n^r .

- For $r < \frac{n}{2}$ the proof of Theorem 5 implies that the only maximum intersecting families are stars.
- For $r = \frac{n}{2}$, let H_0 be a maximal intersecting family of K_{2r}^r : if $E \in H_0$ then $X-E \notin H_0$. If $E \notin H_0$ then there exists an edge $E_j \in H_0$ which does not meet E (by maximality of H_0) thus $X-E = E_j \in H_0$. Thus $|H_0| = \frac{1}{2}m(K_{2r}^r)$. Hence all maximal intersecting families have the same cardinality.

Theorem 6 (Bollobás [1965]). Let $H = (E_1, E_2, \dots, E_m, F_1, F_2, \dots, F_m)$ be a hypergraph of order n with $2m$ edges such that $E_i \cap F_j = \emptyset$ if and only if $i = j$. Then

$$(1) \quad \sum_{j=1}^m \binom{|E_j| + |F_j|}{|E_j|}^{-1} \leq 1.$$

Further, we have equality in (1) if for some integers r, s with $r+s = n$, we have

$$(E_1, E_2, \dots, E_m) = K_n^r; \quad (F_1, F_2, \dots, F_m) = K_n^s.$$

(*) **Proof.** Inspired by an idea of Katona, we may prove the result as follows. Let X be the vertex set of H , and let Y be the set of pairs (S_j, T_j) with $S_j, T_j \subset X$, $S_j, T_j \neq \emptyset$, $S_j \cap T_j = \emptyset$. Form a graph G on Y as follows: two vertices (S_j, T_j) and (S_k, T_k) are adjacent if $S_j \cap T_k = \emptyset$ or $S_k \cap T_j = \emptyset$. Given a permutation π on X and a set $S \subset X$, denote by $\bar{S}$ the smallest interval of the sequence $\sigma = (\pi(1), \pi(2), \dots, \pi(n))$ which contains the set S , and put

$$Y(\pi) = \{(S, T) / (S, T) \in Y; \bar{S} \cap \bar{T} = \emptyset; \bar{S} \text{ is before } \bar{T} \text{ in } \sigma\}.$$

If the vertices (S_j, T_j) and (S_k, T_k) of $Y(\pi)$ are non-adjacent then $\bar{S}_j \cap \bar{T}_j = \emptyset$, $\bar{S}_k \cap \bar{T}_k = \emptyset$, $\bar{S}_j \cap \bar{T}_k \neq \emptyset$, $\bar{S}_k \cap \bar{T}_j \neq \emptyset$ which is a contradiction. Thus $Y(\pi)$ is a clique of G .

Note that if in a graph G on a set Y we consider P cliques $C_1, C_2, \dots, C_p$ and a stable set $S \subset Y$ we obtain, by counting in two different ways the number of pairs (y, C_i) with $y \in S$ and $y \in C_i$,

$$\sum_{y \in S} |\{i / y \in C_i\}| = \sum_{i=1}^p |C_i \cap S| \leq p$$

As the (E_j, F_j) for $j = 1, \dots, m$ constitute a stable set of G we have

$$(2) \quad \sum_{j=1}^m |\{\pi / y(\pi) \in (E_j, F_j)\}| \leq n!$$

Further, for two disjoint sets $E, F \subset X$

$$|\{\pi / (E, F) \in Y(\pi)\}| = \binom{n}{|E \cup F|} (n - |E \cup F|)! |E|! |F|! =$$

$$= n! \binom{|E| + |F|}{|E|}^{-1}$$

This equality, together with (2) gives us relation (1) which was what we had to prove.

4. The coloured edge property and Chvátal's Conjecture

Let $H = (E_1, \dots, E_m)$ be a hypergraph. The *chromatic index* of H is the least number of colours necessary to colour the edges of H such that two intersecting edges are always coloured differently. This number $q(H)$ has been extensively studied for graphs.

If $\Delta_0(H) = k$, then at least k distinct colours are needed to colour the edges of a family of k intersecting edges; thus

$$q(H) \geq \Delta_0(H) \geq \Delta(H).$$

We say that H has the *coloured edge property* if $q(H) = \Delta(H)$, i.e. it is possible to legally colour the edges of H with $\Delta(H)$ colours.

Example 1. Let X be a set of individuals; suppose that certain individuals wish to have meetings during the day, each meeting being defined by a subset E_j of X . We suppose that each individual wishes to attend k meetings. Then we can complete all the reunions in k days if and only if the hypergraph $H = (E_1, E_2, \dots, E_m)$ has the coloured edge property (each colour of an optimal colouring allows us to define the meetings of a day).

Example 2: Bipartite graphs. Let H be a bipartite multigraph defined by a partition (X_1, X_2) of X and some edges E with $|E \cap X_1| = 1$, $|E \cap X_2| = 1$. A well known theorem of König states that H has the coloured edge property.

Example 3: Graphs. Let G be a simple graph, and let $\hat{G}$ be the multigraph obtained from G by adjoining a loop to each vertex. Vizing's theorem says that $q(G) \leq \Delta(G) + 1 = \Delta(\hat{G})$. Then we may colour the edges of $\hat{G}$ with $\Delta(\hat{G})$ colours: this is the coloured edge property.

Example 4: r -complete hypergraphs of order a multiple of r . All complete graphs K_{2p} of order $2p$ even have the coloured edge property; this is an old theorem of Lucas [1892] which he formulated in the following way: a residence of $2p$ girls go for a walk every day in rows of two. Each girl refuses to find herself twice with the same partner. Can you organise the walks for $2p-1$ days? Each of these walks is determined by a colour of the edges of the complete graph K_{2p} . Place the vertices $0, 1, \dots, n-1$ on a circle as in Figure 3, the first colour being determined by the segments of this figure, the others obtained by rotation of the segments about the centre O . In 1936 in Berlin, a student of Schur, R. Pelsesohn, submitted a thesis showing that a school of $3p$ girls can walk every day in rows of 3, that is to say the complete hypergraph K_{3p}^3 has the coloured edge property.

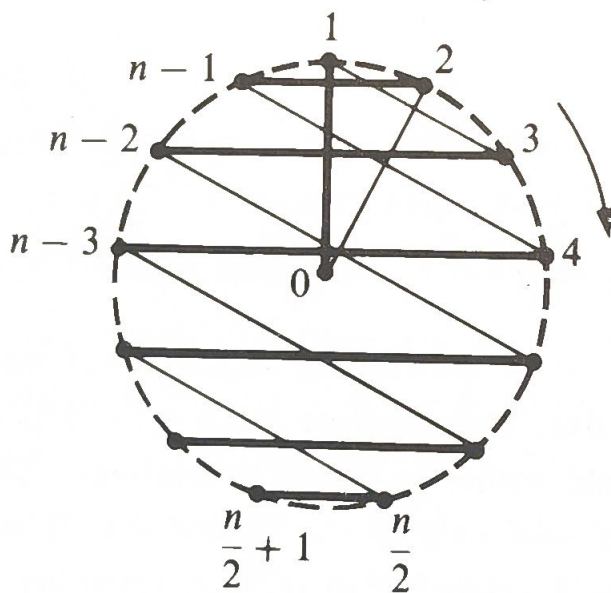


Figure 3

For $p = 3$ this result had been discovered 40 years earlier by Walecki, who had obtained the 28 walks for 9 girls $P, Q, a, b, c, d, e, f, g$ by decomposing the 7 tables shown in Figure 4. Finally, in 1975, Baranyai put a final point on this area of research by showing clearly and simply that K_n^r has the coloured edge property if and only if n is a multiple of r . (For a proof, see §5, Chapter 4).

$$\begin{vmatrix} P & a & b \\ c & d & Q \\ e & f & g \end{vmatrix} \rightarrow \begin{cases} P & a & b, & c & d & Q, & e & f & g \\ P & c & e, & a & d & f, & b & Q & g \\ P & d & g, & c & f & b, & e & a & Q \\ b & d & e, & Q & f & P, & g & a & c \end{cases}$$

$$\begin{vmatrix} P & b & c \\ d & e & Q \\ f & g & a \end{vmatrix}; \begin{vmatrix} P & c & d \\ e & f & Q \\ g & a & b \end{vmatrix}; \begin{vmatrix} P & d & e \\ f & g & Q \\ a & b & c \end{vmatrix}; \begin{vmatrix} P & e & f \\ g & a & Q \\ b & c & d \end{vmatrix}; \begin{vmatrix} P & f & g \\ a & b & Q \\ c & d & e \end{vmatrix}; \begin{vmatrix} P & g & a \\ b & c & Q \\ d & e & f \end{vmatrix}.$$

Figure 4. The seven tables determining the coloring of the edges of K_9^3 .

Example 5. An *interval hypergraph* is a hypergraph whose vertices are points on a line, and each edge is a set of points in an interval. It is easy to see that such a hypergraph has the coloured edge property. This result is also a special case of a more general theorem which we shall prove in Chapter 5.

Let $H = (E_1, E_2, \dots, E_m)$ be a simple hypergraph on X : its *hereditary closure* $\hat{H}$ is the hypergraph on X whose edge set is the set of all non-empty subsets $F \subset X$ such that $F \subset E_i$ for at least one index i .

All families $(F_j/j \in J)$ of non-empty subsets of X such that $F \subset F_j \Rightarrow F = F_k$ for some k are called *hereditary*: clearly we may write this in a unique way as the hereditary closure of a simple hypergraph H .

Not all hereditary hypergraphs satisfy the coloured edge property (e.g. $\hat{K}_7^3$, $\hat{K}_9^4$, $\hat{K}_{10}^4$). Nonetheless, in 1974 Chvátal made the important conjecture:

Chvátal's Conjecture. Every hereditary hypergraph $\hat{H}$ satisfies $\Delta_0(\hat{H}) = \Delta(\hat{H})$.

In other words, in every hereditary hypergraph there is always a star amongst maximum intersecting families. We shall show various cases of this conjecture.

Theorem 7 (Berge [1976]). Let H be a star. Then $\hat{H}$ has the coloured edge property.

Proof. Let H be a simple hypergraph on X , all of whose edges contain a vertex x_0 . Assume the theorem to be true for all hypergraphs having fewer than $m(\hat{H})$ edges. Let A be a maximal subset of X of the form $A = E \cup F$ with $E, F \in \hat{H}$. By the maximality of A , we have $x_0 \in A$. Set

$$\mathcal{B} = \{E/E \in \hat{H}, E \cup F = A \text{ for some } F \in \hat{H}\}$$

1. Observe that $\mathcal{B}$ consists of the sets $E_\lambda \in \mathcal{B}$ with $x_0 \in E_\lambda$ and of the sets of the form $A - E_\lambda$; thus we can colour $\mathcal{B}$ with $d_{\mathcal{B}}(x_0)$ colours, using the same colour for $E_\lambda \in \mathcal{B}$ as for $A - E_\lambda$. Thus if $\hat{H} = \mathcal{B}$ we obtain a colouring of H in a number of colours equal to the degree of x_0 in $\mathcal{B}$ and we are done.

2. Suppose $\hat{H} \neq \mathcal{B}$: we shall show that $\hat{H} - \mathcal{B}$ is an hereditary hypergraph.

Let $E \in \hat{H} - \mathcal{B}$ and $E' \subset E$. Since $E' \in \hat{H}$ it suffices to show that $E' \notin \mathcal{B}$. Otherwise, $E' \cup F' = A$ for an $F' \in \hat{H}$. By maximality of A , we have $E \cup F' = A$; thus $E \in \mathcal{B}$, a contradiction.

3. We now show that the maximal edges of $\hat{H} - \mathcal{B}$ contain x_0 . For, otherwise there exists some $E \in \max(\hat{H} - \mathcal{B})$ with $x_0 \notin E$. Since H is a star, $E \cup \{x_0\} = E_0 \in \hat{H}$. Thus $E_0 \notin \hat{H} - \mathcal{B}$ (by maximality of E); thus $E_0 \in \mathcal{B}$, thus $E_0 \cup F_0 = A$ for some $F_0 \in \hat{H}$. Thus $E \cup (F_0 \cup \{x_0\}) = A$ and $E \in \mathcal{B}$: contradiction.

4. By the induction hypothesis, the edges of $\hat{H} - \mathcal{B}$ can be coloured with $d_{\hat{H}-\mathcal{B}}(x_0)$ colours so, by using part 1 above, we may write:

$$\Delta(\hat{H}) \leq q(\hat{H}) \leq d_{\hat{H}-\mathcal{B}}(x_0) + d_{\mathcal{B}}(x_0) = d_{\hat{H}}(x_0) \leq \Delta(\hat{H})$$

Thus equality holds throughout. This shows that x_0 is a vertex of maximum degree in $\hat{H}$ and that $q(\hat{H}) = \Delta(\hat{H})$.

Q.E.D.

The colouring of the edges of the hereditary closure of K_n^r is related to a well known problem in Operations Research, the "cutting-stock problem", which was solved by Gilmore and Gomory in [1961]; in this problem we wish to cut, from a stock of rods of length n , k_1 poles of length 1, k_2 of length 2, ..., k_r of length r , and to minimise the total number of rods.

Theorem 8 (Baranyai). *Let $r \leq n$ be integers. K_n^r has the coloured edge property if and only if it is possible to solve without waste the cutting stock problem with $k_i = \binom{n}{i}$ for $i = 1, 2, \dots, r$, that is to say, there exists an integer solution (x_j^i) to the system*

$$x_j^i \geq 0,$$

x_j^i is the number of i -subsets to colour with j ,

$$\sum_{i=1}^r ix_j^i = n \quad (j = 1, 2, \dots)$$

$$\sum_j x_j^i = \binom{n}{i} \quad (i = 1, 2, \dots, r).$$

It is clear that this condition is necessary; it is also sufficient, as we shall show later (Corollary to Baranyai's Theorem, §5, Chapter 4).

Just as the r -complete hypergraph K_n^r generalises the complete graph, we may generalise the complete bipartite graph by the r -partite complete hypergraph $K_{n_1, n_2, \dots, n_r}^r$ defined as follows: let $X^1, X^2, \dots, X^r$ be disjoint sets with $|X^i| = n_i$ for $i = 1, 2, \dots, r$.

The vertices are the elements of $X^1 \cup X^2 \cup \dots \cup X^r$, and the edges are all sets of the form $\{x^1, x^2, \dots, x^r\}$ with $x^1 \in X^1, x^2 \in X^2, \dots, x^r \in X^r$.

Theorem 9 (Berge, Johnson [1977]). *The r -partite complete hypergraph $K_{n_1, n_2, \dots, n_r}^r$ and its hereditary closure have the coloured edge property.*

Proof.

1. Let $H = K_{n_1, n_2, \dots, n_r}^r$, with $1 \leq n_1 \leq n_2 \leq \dots \leq n_r$, $r \geq 2$. We shall show that we can colour the edges of H with $\Delta(H) = n_2 n_3 \dots n_r$ colours. We denote the elements of X^k by $x_1^k = 0, x_2^k = 1, \dots, x_{n_k}^k = n_k - 1$. As usual, denote by $[p]_k$ the integer $\leq k-1$ congruent to p modulo k . Associate with each edge $\bar{x} = x^1 x^2 \dots x^r$ of H the $(r-1)$ -tuple

$$\xi(\bar{x}) = ([x^2 + x^1]_{n_2}, [x^3 + x^1]_{n_3}, \dots, [x^r + x^1]_{n_r})$$

If two distinct edges $\bar{x} = x^1 x^2 \dots x^r$ and $\bar{y} = y^1 y^2 \dots y^r$ intersect, then one of the two following cases occurs:

(i) $x^1 = y^1$ and then there is an $i \geq 2$ with $x^i \neq y^i$, so

$$[x^i + x^1]_{n_i} \neq [y^i + y^1]_{n_i} \text{ and } \xi(\bar{x}) \neq \xi(\bar{y});$$

(ii) $x^1 \neq y^1$ and then there is an $i \geq 2$ with $x^i = y^i$, so

$$[x^i + x^1]_{n_i} \neq [y^i + y^1]_{n_i} \text{ and } \xi(\bar{x}) \neq \xi(\bar{y}).$$

We may consider the map $\bar{x} \rightarrow \xi(\bar{x})$ as a colouring of the edges, and the number of distinct colours used is at most $n_2 n_3 \dots n_r = \Delta(K_{n_1, n_2, \dots, n_r}^r)$.

Q.E.D.

2. We shall show that $\hat{H}$ can be coloured with $\Delta(\hat{H})$ colours. For $i = 1, 2, \dots, n$, consider an additional vertex a^i , and put $Y^i = X^i \cup \{a^i\}$; consider the hypergraph $H' = K_{n_1+1, n_2+1, \dots, n_r+1}^r$ determined by the classes Y^i , $|Y^i| = n_i + 1$.

For each edge E of $\hat{H}$ there is an edge F of H' defined by $F = E \cup \{a^i / E \cap X^i = \emptyset\}$.

Thus there is a bijection between the edges of $\hat{H}$ and those of H' . As we have shown that the r -partite complete hypergraph has the coloured edge property, we can colour the edges of H' with

$$\Delta(H') = (n_2+1)(n_3+1)\dots(n_r+1)$$

colours. If we colour each edge E of $\hat{H}$ with the colour of the corresponding edge F of H' , it is clear that two edges of $\hat{H}$ which intersect have different colours. Hence

$$q(\hat{H}) \leq q(H') = \Delta(H') = \Delta(\hat{H}) \leq q(\hat{H}).$$

Thus $q(\hat{H}) = \Delta(\hat{H})$ and the hypergraph $\hat{H}$ has the coloured edge property.

Q.E.D.

The main hypergraphs H for which it has been shown that $\Delta_0(\hat{H}) = \Delta(\hat{H})$ are the following:

1. H is a star (Schönheim [1973]). In this case, Theorem 7 shows that $\hat{H}$ has a stronger property, the coloured edge property.

2. H is 2-uniform (Vizing).

3. H is 3-uniform (Sterboul [1974]). In this case it can also be shown that the maximum intersecting families of $\hat{H}$ have one of the following structures:

- $\hat{H}(a)$ (star);
- $\{ab, ac, bc, abc\}$;
- $\{ab, ac, ad, abc, abd, acd, bcd\}$
- $\{abx_1, abx_2, \dots, abx_p, acx_1, \dots, acx_p, bcx_1, \dots, bcx_p, ab, ac, bc, abc\}$.

4. H is linear.

If H is uniform, see Sterboul [1974];

For all H , see Stein [1983].

5. H is of degree $\Delta(H) = 2$ (Stein, Schönheim [1978], Wang and Wang [1983]).

6. H is an r -partite complete hypergraph. In this case Theorem 9 shows that $\hat{H}$ has the stronger coloured edge property.

7. H is the complete hypergraph K_n^r with $r \leq \frac{n}{2}$ (from Theorem 5).

Example 4 suggests the following conjecture:

Conjecture. *If H is linear then $\hat{H}$ has the coloured edge property.*

This conjecture is true if H is a graph (Vizing); if H is a projective plane on 7 points 1,2,...,7, we can colour the edges of $\hat{H}$ with $\Delta(\hat{H}) = 10$ colours in the following way:

colour 1: 123, 45, 6, 7	colour 6: 345, 12, 67
colour 2: 147, 56, 23	colour 7: 367, 14, 25
colour 3: 156, 34, 27	colour 8: 17, 36, 24, 5
colour 4: 246, 37, 15	colour 9: 16, 35, 47, 2
colour 5: 257, 13, 46	colour 10: 57, 26, 1, 3, 4.

5. The Helly property

Let $H = (E_1, E_2, \dots, E_m)$ be a simple hypergraph. We say that H has the *Helly property* if every intersecting family of H is a star, i.e. for $J \subset \{1, 2, \dots, m\}$,

$$E_j \cap E_k \neq \emptyset \quad (j, k \in J)$$

implies

$$\bigcap_{j \in J} E_j \neq \emptyset.$$

Hence a graph has the Helly property if and only if it is triangle-free; hypergraphs with the Helly property have also other properties which generalise those of triangle free graphs.

Example 1. Let H be an interval hypergraph: its vertices are points on a line, and its edges are intervals of points. A theorem of Helly shows that H has the Helly property.

Example 2 (Algebra). Let $(X, \leq)$ be a lattice, i.e. an ordered set such that for each pair (a, b) there exists a least upper bound $a \vee b$ and a greatest lower bound $a \wedge b$.

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Let H be a family of intervals of the form

$$E(a,b) = \{x/a \leq x \leq b\}.$$

Then it can be shown that H has the Helly property. If X is the set of natural numbers, and if the edges of H are arithmetic progressions, the Helly property is known as the "Chinese Remainder Theorem" (cf. Ore [1952]).

We shall say that a hypergraph $H = (E_1, E_2, \dots, E_m)$ is k -Helly if for every set $J \subset \{1, 2, \dots, m\}$, the following two conditions are equivalent:

$$(D_k) \quad I \subset J, |I| \leq k, \text{ implies } \bigcap_{i \in I} E_i \neq \emptyset;$$

$$(D) \quad \bigcap_{j \in J} E_j \neq \emptyset$$

Clearly if J satisfies (D) then it also satisfies (D_k) ; if H is not k -Helly there are also sets J which satisfy (D_k) but not (D) .

Clearly, a hypergraph is 2-Helly if and only if it satisfies the Helly property. Note also that if a hypergraph is k -Helly, we have $(D_{k+1}) \Rightarrow (D_k) \Rightarrow (D)$; thus a $(k+1)$ -Helly hypergraph is also k -Helly.

Example. Let H be a hypergraph such that if each vertex is a point of $\mathbb{R}^d$ and each edge is the set of points contained in a convex set: an interval hypergraph corresponds to the case $d = 1$. A theorem of Helly states that such a hypergraph in $\mathbb{R}^d$ is $(d+1)$ -Helly.

Theorem 10 (Berge, Duchet [1975]). A hypergraph H is k -Helly if and only if for every set A of vertices with $|A| = k+1$, the intersection of the edges E_j with $|E_j \cap A| \geq k$ is non-empty.

Proof.

1. Let H be a k -Helly hypergraph on X ; let A be a subset of X with $|A| = k+1$. Set

$$J = \{j / |E_j \cap A| \geq k\}.$$

We shall show that $\bigcap_{j \in J} E_j \neq \emptyset$.

Case 1. $|J| \leq k$. We have $\bigcap_{j \in J} E_j \neq \emptyset$ since otherwise the bipartite incidence graph G of the vertices of A versus the edges $(E_j/j \in J)$ satisfies

$$|J|k \leq \sum_{j \in J} d_G(j) = m(G) \leq (|J|-1)|A| = (|J|-1)(k+1)$$

which implies $|J| \geq k+1$: a contradiction.

Case 2. $|J| \geq k+1$. In this case each set $I \subset J$ with $|I| \leq k$ satisfies $\bigcap_{i \in I} E_i \neq \emptyset$ (from Case 1); thus J satisfies (D_k) and hence (D) . Thus

$$\bigcap_{j \in J} E_j \neq \emptyset.$$

2. Let $H = (E_1, E_2, \dots, E_m)$ be a hypergraph such that for each $A \subset X$ with $|A| = k+1$, the family $(E_j/|E_j \cap A| \geq k)$ has a non-empty intersection. We shall show that H is k -Helly, that is for every $J \subset \{1, 2, \dots, m\}$, $(D_k) \Rightarrow (D)$.

The proof is by induction on $|J|$. Clearly this is true for $|J| \leq k$, so assume $|J| > k$; let $j_1, j_2, \dots, j_{k+1}$ be distinct elements of J . Then the condition (D_k) implies

$$(\forall I \subset J - \{j_\lambda\}, |I| \leq k): \bigcap_{i \in I} E_i \neq \emptyset$$

By the induction hypothesis this implies

$$\bigcap_{j \in J - \{j_\lambda\}} E_j \neq \emptyset.$$

Let a_λ be an element in this intersection. The elements $a_1, a_2, \dots, a_{k+1}$ are different (otherwise we are done). For $A = \{a_1, a_2, \dots, a_{k+1}\}$,

$$|E_j \cap A| \geq k \quad (j \in J)$$

whence

$$\bigcap_{j \in J} E_j \neq \emptyset.$$

Q.E.D.

Corollary. *A hypergraph H has the Helly property if and only if for any three vertices a_1, a_2, a_3 , the family of edges containing at least two of the vertices a_i has a non-empty intersection.*

Application: Family of subtrees of a tree. Let G be an acyclic connected graph on X , i.e. G is a tree. Consider a family H of subsets of X which induce a subtree of

G . We shall show, with the help of the preceding corollary, that H has the Helly property. To see this, consider three vertices a, b, c of G . If $\mu[x, y]$ denotes the unique path in the tree G connecting the vertices x and y it is easy to see that the three paths $\mu[a, b]$, $\mu[b, c]$ and $\mu[c, a]$ have a common vertex x_0 (otherwise G would have a cycle). This vertex x_0 belongs to every edge of H containing two of the points a, b, c . Thus H has the Helly property.

(Note that if G is a path P_n we obtain Helly's Theorem).

Theorem 11 (Tuza [1984]). *Let $H = (E_1, E_2, \dots, E_m)$ be a simple k -Helly hypergraph of order n . If $\min_j |E_j| \geq k+1$ then*

$$\sum_{j=1}^m \binom{n-1}{|E_j|-1}^{-1} \leq 1$$

(*) **Proof.**

1. We shall show first that every edge E_j contains a vertex a_j such that $E_j - \{a_j\}$ is not contained in any edge other than E_j . Indeed, if this is not the case, there exists an edge $E_0 = \{a_1, a_2, \dots, a_r\}$ with $r \geq k+1$, such that, say, $E_0 - \{a_i\} \subset E_i$ for $i = 1, 2, \dots, r$. Since H is a simple hypergraph, we have $E_0 \cap E_i = E_0 - \{a_i\}$ for $i = 1, 2, \dots, r$. Thus

$$\bigcap_{j=0}^r E_j = \emptyset.$$

However, the intersection of $r-1$ of the sets $E_0, E_1, \dots, E_r$ is non-empty. Since $r-1 \geq k$, and since H is k -Helly, we have also:

$$\bigcap_{j=1}^r E_j \neq \emptyset.$$

A contradiction follows.

2. Thus every edge E_j contains a vertex a_j such that

$$(E_j - \{a_j\}) \cap (X - E_i) \neq \emptyset \quad \text{for all } i \neq j$$

Set

$$E'_j = E_j - \{a_j\}$$

$$F'_j = X - E_j.$$

Thus we have

$$E_j' \cap F_j' = \emptyset$$

$$E_j' \cap F_i' \neq \emptyset \text{ if } i \neq j$$

We may now apply Theorem 6 and

$$\sum_{j=1}^m \binom{|X-E_j|+|E_j|-1}{|E_j|-1} \leq 1,$$

The theorem follows.

Corollary (Bollobás, Duchet [1979]). *Let H be a simple k -Helly hypergraph of order n with $\min_j |E_j| \geq k+2$ and $\max_j |E_j| = r \leq \frac{n}{2}$. Then*

$$(1) \quad m(H) \leq \binom{n-1}{r-1}.$$

Proof. Every $E \in H$ satisfies:

$$\binom{n-1}{|E|-1} \leq \binom{n-1}{r-1}.$$

Hence

$$m(H) \binom{n-1}{r-1}^{-1} \leq \sum_{E \in H} \binom{n-1}{|E|-1}^{-1} \leq 1.$$

Inequality (1) follows.

For a hypergraph H with the Helly property, more precise results can be proved:

Theorem 13 (Bollobás, Duchet [1983]). *Let H be a simple hypergraph of rank $r \geq 3$, $r \leq \frac{n}{2}$, with the Helly property. Then*

$$(1) \quad m(H) \leq \binom{n-1}{r-1}.$$

Further, equality holds in (1) if and only if H is a star of K_n^r .

Theorem 14 (Bollobás, Duchet [1983]). *Let H be a simple hypergraph of order $n \geq 5$ with the Helly property. Then*

$$(1') \quad m(H) \leq \binom{n-1}{\lfloor n/2 \rfloor}.$$

Further, equality holds in (1') if and only if one of the following is true:

- (i) $n = 2h$ is even and H is a star of K_n^h ;
- (ii) $n = 2h+1$ is odd ≥ 7 and H is a star of K_n^{h+1} ;
- (iii) $n = 5$ and H is a star of K_5^3 , or is the bipartite complete graph $K_{2,3}$ with one class of 2 vertices and one of 3 vertices.

6. Section of a hypergraph and the Kruskal-Katona Theorem

Let H be a simple hypergraph on X of rank r , and let $k \leq r$ be a positive integer. Define the k -section of H to be a hypergraph $[H]_k$ whose edges are the sets $F \subset X$ satisfying either $|F| = k$, and $F \subseteq E$ for some $E \in H$; or $|F| < k$ and $F = E$ for some $E \in H$.

Observe that $[H]_k$ is a simple hypergraph on X . Further its rank is k .

For $k = 2$, the 2-section $[H]_2$ is thus a graph; if H contains no loops then $[H]_2$ is a simple graph which is obtained by joining two vertices of X if they belong to the same edge of H . If H is a simple r -uniform hypergraph with m edges, what can we say about the number of edges of $[H]_{r-1}$?

The best possible lower bounds for all m were obtained independently by Kruskal [1963] and Katona [1968]. The proof was simplified by Daykin [1976], and that which we now give, shorter still, is due to Frankl [1984]. We need two preliminary lemmas.

Lemma 1. *Let m and r be positive integers. Then there exist integers $a_r, a_{r-1}, \dots, a_s$ such that*

$$(1) \quad m = \binom{a_r}{r} + \binom{a_{r-1}}{r-1} + \dots + \binom{a_s}{s}$$

$$(2) \quad a_r > a_{r-1} > \dots > a_s \geq s \geq 1.$$

Further the a_i 's are defined uniquely by (1) and (2). In particular, a_r is the largest integer such that

$$m - \binom{a_r}{r} \geq 0.$$

(*) **Proof** (by induction on r). For $r = 1$, we have $1 = r \geq s \geq 1$ so $s = 1$ and $a_s = m$; thus the decomposition (1) exists and is unique. Assume now the existence and uniqueness of decomposition (1) for $r-1$. Let a_r be the largest integer such that $m - \binom{a_r}{r} \geq 0$. Then by the induction hypothesis,

$$m - \binom{a_r}{r} = \binom{a_{r-1}}{r-1} + \cdots + \binom{a_s}{s}$$

$$a_{r-1} > a_{r-2} > \cdots > a_s \geq s.$$

We must have $a_r > a_{r-1}$ since otherwise we could write

$$m \geq \binom{a_r}{r} + \binom{a_{r-1}}{r-1} \geq \binom{a_r}{r} + \binom{a_r}{r-1} = \binom{a_r+1}{r}.$$

This contradicts the definition of a_r . Hence the existence of decomposition (1) is proven.

To show uniqueness, suppose there exist two distinct decompositions of m :

$$m = \binom{a_r}{r} + \cdots + \binom{a_s}{s} = \binom{b_r}{r} + \cdots + \binom{b_s}{s}$$


Observe that

$$m \leq \binom{a_r}{r} + \binom{a_{r-1}}{r-1} + \cdots + \binom{a_{r-r+1}}{r-r+1} = \binom{a_r+1}{r}$$

If $a_r < b_r$ then

$$m \leq \binom{a_r+1}{r} \leq \binom{b_r}{r} \leq m$$

This implies $m = \binom{a_r+1}{r}$ contradicting the definition of r .

Hence $a_r = b_r$, and as the decomposition of $m - \binom{a_r}{r} = m - \binom{b_r}{r}$ is unique (induction hypothesis) the two decompositions of m are identical.

Lemma 2 (Frankl [1984]). Let H be an r -uniform hypergraph on X and let $x_1 \in X$. There exists an r -uniform hypergraph H' on X with $m(H') = m(H)$, $m([H']_{r-1}) \leq m([H]_{r-1})$ and satisfying

$$F \in [H' - H'(x_1)]_{r-1} \Rightarrow F \cup \{x_1\} \in H'.$$

(*) **Proof.** For a vertex $x \neq x_1$, put

$$\sigma_x E = \begin{cases} (E - \{x\}) \cup \{x_1\} & \text{if } x \in E, x_1 \notin E \\ E & \text{otherwise} \end{cases}$$

Put $\sigma_x H = \{\sigma_x E / E \in H\}$. It is easy to see that $[\sigma_x H]_{r-1} \subset \sigma_x [H]_{r-1}$. By repeating the operation σ_y on $\sigma_x H$ as many times as necessary we get a hypergraph H' with $m(H') = m(H)$, $m([H']_{r-1}) \leq m([H]_r)$, and $\sigma_x H' = H'$ for all $x \neq x_1$.

Theorem 14 (Kruskal, Katona). Let H be an r -uniform hypergraph with

$$m(H) = m = \binom{a_r}{r} + \binom{a_{r-1}}{r-1} + \cdots + \binom{a_s}{s}$$

$$a_r > a_{r-1} > \cdots > a_s \geq s \geq 1.$$

Then

$$m([H]_{r-1}) \geq \binom{a_r}{r-1} + \binom{a_{r-1}}{r-2} + \cdots + \binom{a_s}{s-1}$$

(*) **Proof** (by induction on r and m).

1. We may assume that H satisfies

$$(1) \quad F \in [H - H(x_1)]_{r-1} \Rightarrow F \cup \{x_1\} \in H$$

(simply by replacing H by the hypergraph H' defined in Lemma 2). Set $H_1 = (E - \{x_1\} / E \in H(x_1))$. Then

$$(2) \quad m([H]_{r-1}) \geq m(H_1) + m([H_1]_{r-2}).$$

2. The theorem holds trivially for $r = 1$ or $m = 1$; proceed now by induction on r and on m .

Suppose first $a_r - 1$

$$(3) \quad m(H_1) \geq \binom{a_{r-1}}{r-1} + \cdots + \binom{a_s-1}{s-1}$$

By applying the induction hypothesis to the hypergraph H_1 (less some edges if the inequality is strict), we obtain

$$m([H_1]_{r-2}) \geq \binom{a_r-1}{r-2} + \cdots + \binom{a_s-1}{s-2}$$

Thus, from (2),

$$\begin{aligned} m([H]_{r-1}) &\geq m(H_1) + m([H_1]_{r-2}) \\ &\geq \binom{a_r-1}{r-1} + \cdots + \binom{a_s-1}{s-1} + \binom{a_r-1}{r-2} + \cdots + \binom{a_s-1}{s-2} \\ &= \binom{a_r}{r-1} + \cdots + \binom{a_s}{s-1} \end{aligned}$$

which is what we had to show.

Suppose now that

$$(4) \quad m(H_1) < \binom{a_r-1}{r-1} + \binom{a_{r-1}-1}{r-2} + \cdots + \binom{a_s-1}{s-1}$$

As a consequence we can write

$$\begin{aligned} m(H-H(x_1)) &= m(H) - m(H_1) > \binom{a_r}{r} + \cdots + \binom{a_s}{s} - \binom{a_r-1}{r-1} - \cdots - \binom{a_s-1}{s-1} \\ &= \binom{a_r-1}{r-1} + \binom{a_{r-1}-1}{r-2} + \cdots + \binom{a_s-1}{s} \end{aligned}$$

From (1), and applying the induction hypothesis on m to $H-H(x_1)$,

$$m(H_1) \geq m([H-H(x_1)]_{r-1}) \geq \binom{a_r-1}{r-1} + \binom{a_{r-1}-1}{r-2} + \cdots + \binom{a_s-1}{s-1}$$

which contradicts (4).

Corollary. Let H be an r -uniform hypergraph and let k be an integer with $r > k \geq 2$. If a is the largest integer such that $m(H) \geq \binom{a}{r}$ then

$$m([H]_k) \geq \binom{a}{k}$$

Proof. Let H_1 be a partial hypergraph of H with $m(H_1) = \binom{a}{r}$. From Theorem 14,

$$m([H_1]_{r-1}) \geq \binom{a}{r-1}$$

Let H_2 be a partial hypergraph of $[H_1]_{r-1}$ with $m(H_2) = \binom{a}{r-1}$. By Theorem 14,

$$m([H_2]_{r-2}) \geq \binom{a}{r-2},$$

etc. Finally, $m([H_{r-k}]_k) \geq \binom{a}{k}$. Since $[H]_k \supset [H_{r-k}]_k$ we also have

$$m([H]_k) \geq \binom{a}{k}.$$

Q.E.D.

7. Conformal Hypergraphs

We say that a hypergraph H is *conformal* if all the maximal cliques of the graph $[H]_2$ are edges of H . If H is simple, it is conformal if and only if the edges of H are the maximal cliques of a graph.

More generally, consider an integer $k \geq 2$. Every edge A of a hypergraph H satisfies the property: the edges of $[H]_k$ contained in A constitute a k -complete hypergraph. If every set $A \subset X$ maximal with this property is an edge of H the hypergraph is said to be *k-conformal*. Hence a hypergraph is conformal if and only if it is 2-conformal.

Proposition. *A hypergraph H is k -conformal if and only if for every set $A \subset X$ the following two conditions are equivalent:*

- (C_k) every $S \subset A$ with $|S| \leq k$ is contained in some edge of H ,
- (C) the set A is contained in an edge of H .

Observe that (C) always implies (C_k).

Lemma. *A hypergraph is k -conformal if and only if its dual is k -Helly.*

Proof. In the hypergraph $H = (E_1, E_2, \dots, E_m)$ the set $A = \{x_j / j \in J\}$ satisfies the condition (C_k) if and only if in the dual hypergraph $H^* = (X_1, X_2, \dots, X_n)$ the set J satisfies

$$(D_k) \quad I \subset J, |I| \leq k \text{ implies } \bigcap_{i \in I} X_i \neq \emptyset$$

Similarly, the set A satisfies Condition (C) if and only if in the dual hypergraph H^* , the set J satisfies

$$(D) \quad \bigcap_{j \in J} X_j \neq \emptyset.$$

Thus (C_k) is equivalent to (C) if and only if (D_k) is equivalent to (D).

Theorem 15. *A simple hypergraph H is k -conformal if and only if for each partial hypergraph $H' \subset H$ having $k+1$ edges, the set $\{x/x \in X, d_{H'}(x) \geq k\}$ is contained in an edge of H .*

Proof. From Theorem 10, the dual hypergraph $H^* = (X_1, X_2, \dots, X_n)$ is k -Helly if and only if for a set $F = \{e_j/j \in J\}$ with $|J| = k+1$, the intersection of the X_i with $|X_i \cap J| \geq k$ is non-empty. Or, again, for each $H' = \{E_j/j \in F\}$ with $|J| = k+1$ there exists an edge of H which contains the set

$$\{x/d_{H'}(x) \geq k\}.$$

Corollary (Gilmore's Theorem). *A necessary and sufficient condition for a hypergraph H to be conformal is that for any three edges E_1, E_2, E_3 , the hypergraph H has an edge containing the set*

$$(E_1 \cap E_2) \cup (E_1 \cap E_3) \cup (E_2 \cap E_3)$$

It suffices to put $k = 2$ in the statement of Theorem 15.

8. Representative Graphs

Given a hypergraph $H = (E_1, E_2, \dots, E_m)$ on X , its *representative graph*, or *line-graph* $L(H)$ is a graph whose vertices are points $e_1, e_2, \dots, e_m$ representing the edges of H , the vertices e_i, e_j being adjacent if and only if $E_i \cap E_j \neq \emptyset$.

Example 1. The representative graph of a simple graph G was characterised by Beineke [1968]: a graph is an $L(G)$ if and only if it does not contain as an induced sub-graph any of the graphs $G_1, G_2, \dots, G_9$ shown in Figure 5.

Example 2. The representative graph of a multigraph G was characterised by Bermond and Meyer [1973]: a graph is an $L(G)$ if and only if it does not contain any of the graphs $G'_1, G'_2, \dots, G'_7$ shown in Figure 6.

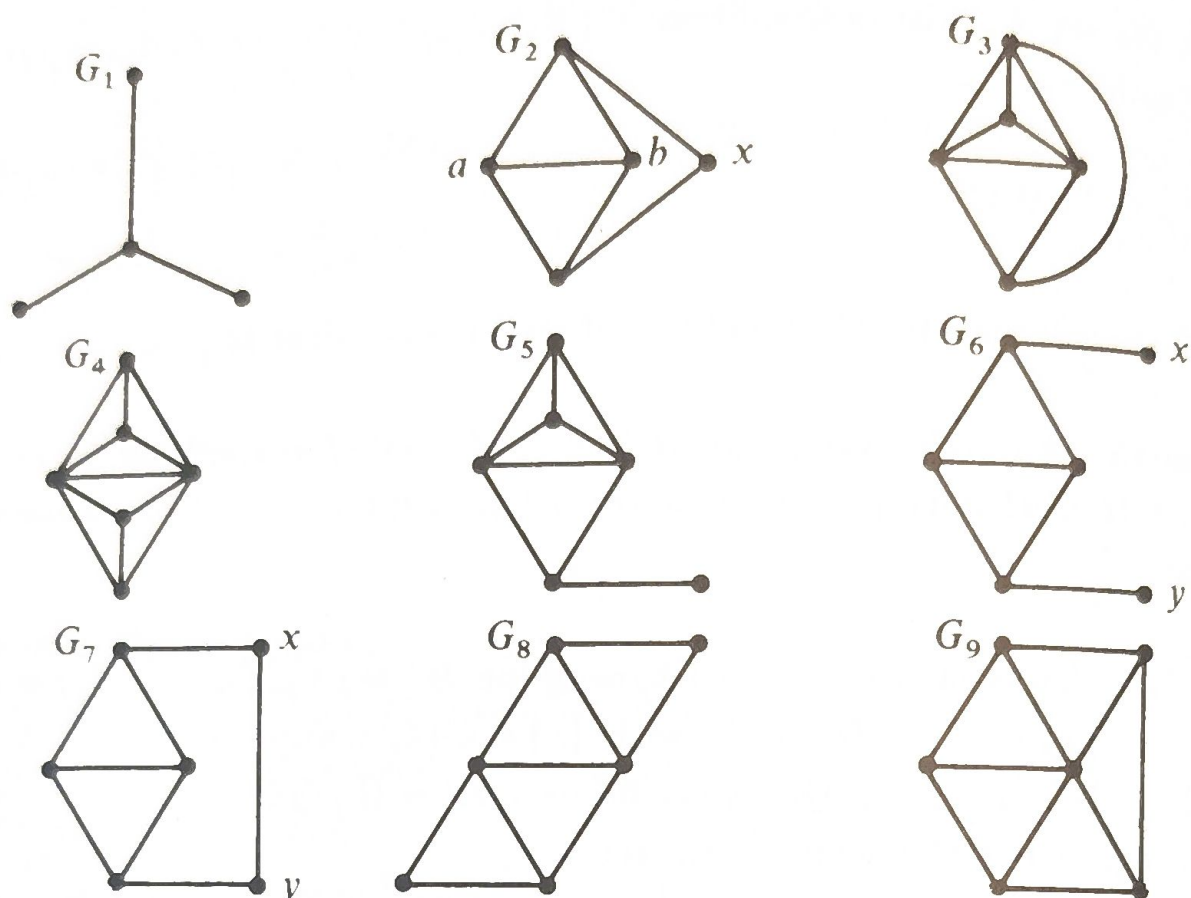


Figure 5. The 9 forbidden configurations for the representative graph of a simple graph.

Example 3. The representative graph of a multigraph without triangles or loops is characterised by: each vertex appears in at most 2 maximal cliques.

Example 4. The representative graph of a bipartite multigraph is characterised by: each vertex appears in at most two maximal cliques, and every elementary odd cycle contains two sides of a triangle.

Example 5. If H is a family of intervals on a line, there is a characterisation of $L(H)$ due to Gilmore and Hoffman (cf. *Graphs*, Chapter 16, Theorem 12): it is a triangulated complement of a comparability graph. This concept has a simple interpretation: if m individuals were present during various intervals of time in a meeting room, a detective who demands of each person whom he has met can trace the "graph of meetings": if nobody lies, the graph represents a family of intervals.

We do not know any similar characterisation for the representative graph of a family of convex sets in the plane, but we do know that every graph represents convex sets in the 3-dimensional space (Wegner [1965]).

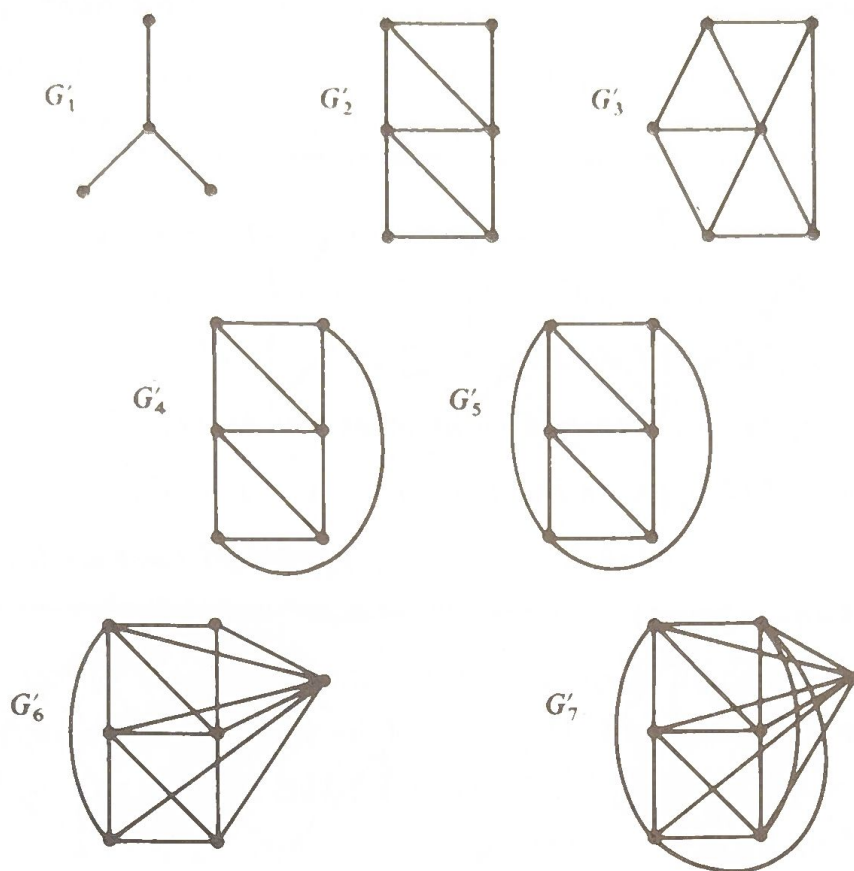


Figure 6. The 7 forbidden configurations for the representative graph of a multi-graph.

Proposition 1. The representative graph of a hypergraph H is the 2-section $[H^*]_2$. Further, the following two properties are equivalent:

- (i) H satisfies the Helly property and G is the representative graph of H ;
- (ii) the maximal edges of H^* are the maximal cliques of G .

Clearly the graph $[H^*]_2$ is isomorphic to $L(H)$, but $[H^*]_2$ can have loops if H^* has loops.

For the other part, if H has the Helly property, H^* is conformal; thus (i) implies that $G = [H^*]_2$ has as cliques the maximal edges of H^* . Similarly (ii) implies (i).

Observe that if $G = L(H)$ and if H does not satisfy the Helly property it can happen that H^* does not contain the maximal cliques of $L(H)$. For example, if H is the hypergraph H_2 in Figure 8, $L(H)$ is the graph G in Figure 8; the maximal cliques of G are not the edges of H^* .

Proposition 2. *Every graph is the representative graph of a linear hypergraph.*

A simple graph G on $\{x_1, \dots, x_n\}$ is the representative graph of a linear hypergraph $(X_1, X_2, \dots, X_n)$ if we take for X_i the set of edges of G adjacent to the vertex x_i .

Proposition 3. *A graph G is the representative graph of an r -uniform hypergraph if and only if G contains a family $\mathcal{C}$ of cliques with the following properties:*

- (Π_0) each clique of $\mathcal{C}$ is of cardinality ≥ 2 ;
- (Π_1) every edge of G is contained in at least one clique of $\mathcal{C}$;
- (Π_2) each vertex of G appears in at most r cliques of $\mathcal{C}$;
- (Π_3) for each vertex x which is covered by exactly r cliques of $\mathcal{C}$, the intersection of these cliques is $\{x\}$.

Indeed, consider the r -regular hypergraph $\mathcal{C}'$ obtained from $\mathcal{C}$ by adjoining loops, which is always possible because of (Π_2) . Let H be the dual of the hypergraph $\mathcal{C}'$. By (Π_1) we have $L(H) = [H^*]_2 = [\mathcal{C}] = G$. By (Π_3) the hypergraph H has no repeated edges: it is thus an r -uniform hypergraph.

Proposition 4. *A graph G is the representative graph of a linear r -uniform hypergraph if and only if, in G there exists a family $\mathcal{C}$ of cliques satisfying (Π_0) , (Π_2) and*

- (Π'_1) each edge is contained in exactly one clique of $\mathcal{C}$.

Let $\mathcal{C}'$ be the r -regular hypergraph obtained from $\mathcal{C}$ by adding loops, which is possible from (Π_2) . Let H be the dual hypergraph of $\mathcal{C}'$. From (Π'_1) , $L(H) = G$, the hypergraph $\mathcal{C}$ is linear and hence its dual H is linear (Proposition 3, §3).

One can ask if it is possible to characterise $L(H)$ by a finite family of forbidden subgraphs in the case $r \neq 2$. In fact, Nickel, then Gardner, then Bermond, Germa, Sotteau [1977] exhibited an infinite family of forbidden configurations for a representative graph of a 3-uniform hypergraph.

The graphs $G_1(t)$, $G_2(t)$, $G_3(t)$ of Figure 7 constitute infinite families of minimal excluded configurations for the representative graph of a 3-uniform linear hypergraph.

Nonetheless, it can be shown that

Theorem 16 (Naik, Rao, Shrikhande, Singhi [1982]). *There exists a finite family $\mathcal{F}_3$ of graphs such that every graph G with minimum degree ≥ 69 is the representative*

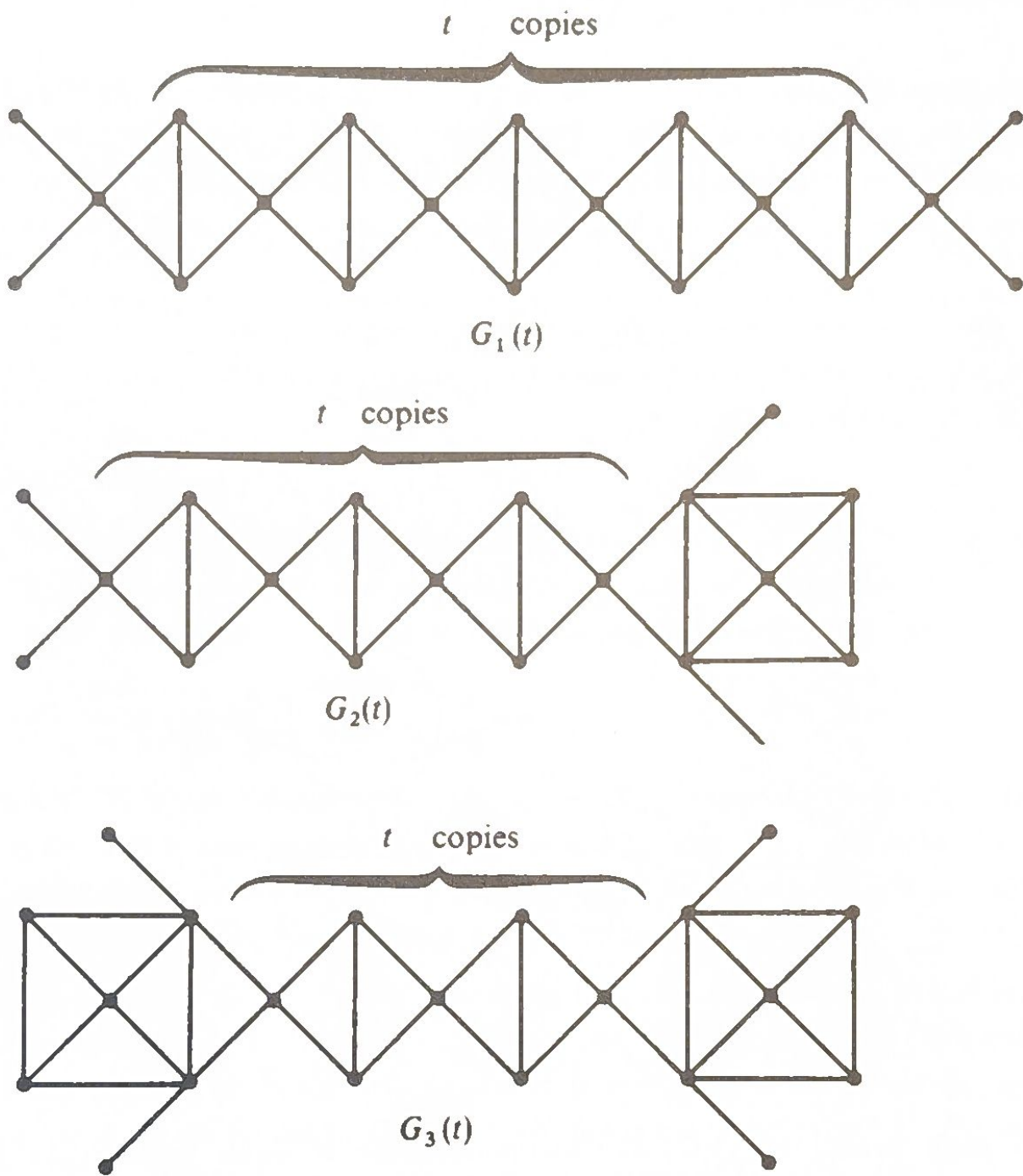


Figure 7

graph of a linear 3-uniform hypergraph if and only if G contains no member of $\mathcal{F}_3$ as an induced subgraph.

More generally, they show the existence of a cubic polynomial $f(k)$ with the property that for each k there exists a finite family $\mathcal{F}_k$ of forbidden graphs such that every graph G of minimum degree $\geq f(k)$ is the representative graph of a linear k -uniform hypergraph if and only if G does not contain a member of $\mathcal{F}_k$ as an induced subgraph.

By way of example, we can check the preceding propositions on the graph G of Figure 8 which is, at one and the same time, the representative graph of the hypergraphs H_1 , H_2 and H_3 of Figure 8.

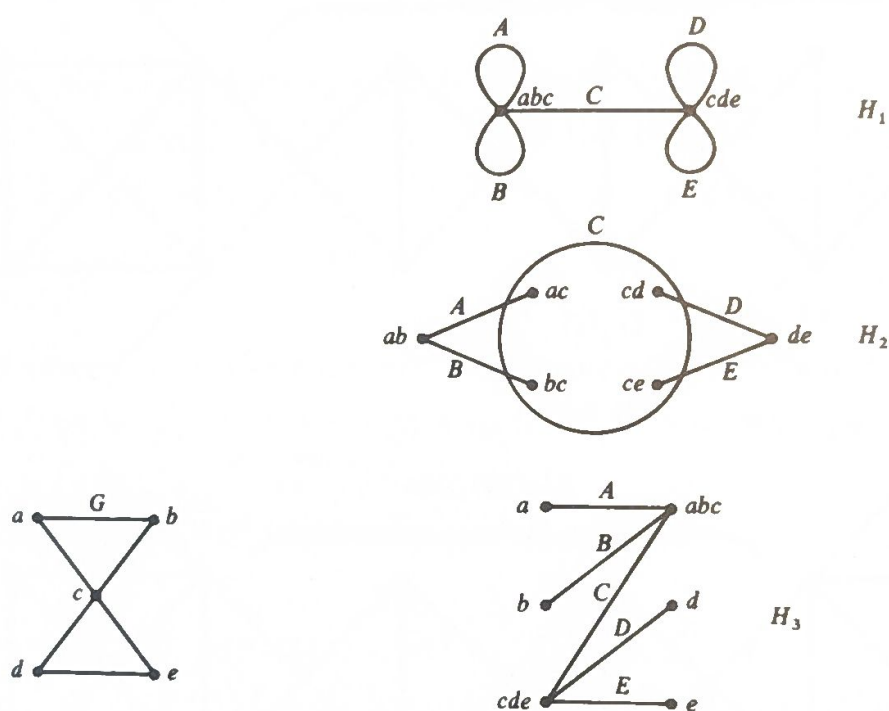


Figure 8

We shall denote by $\Omega(G)$ the minimum order of those hypergraphs H with $G = L(H)$; for example, for the graph G in Figure 8, $\Omega(G) = 2$ since $G = L(H_1)$.

The determination of $\Omega(G)$ brings us back to the determination of the chromatic number by the following result.

Lemma. Let G be a graph on $\{x_1, x_2, \dots, x_n\}$ without isolated vertices, and let $\bar{G}$ be the graph whose vertices correspond to the edges of G , the vertices corresponding to the edges $[a,b]$ and $[x,y]$ of G being adjacent if and only if $\{a,b,x,y\}$ is not a clique in G (i.e. at least one of ax, ay, bx, by is not an edge of G). Then the minimum order $\Omega(G)$ of the hypergraphs for which G is the representative graph is equal to the chromatic

number of $\bar{G}$.

Proof.

1. We shall show that to each q -colouring $(\bar{S}_1, \dots, \bar{S}_q)$ of the vertices of $\bar{G}$ with q colours we may associate a hypergraph $H = (X_1, X_2, \dots, X_n)$ of order q such that $G = L(H)$.

Indeed, the set $\bar{S}_i$ of vertices of $\bar{G}$ coloured with colour i is stable; if $[a, b]$ is an edge of G belonging to $\bar{S}_i$, the vertex a is adjacent to each end of any edge in $\bar{S}_i$. The ends of the edges of $\bar{S}_i$ thus generate a clique E_i of G . The hypergraph $\mathcal{C} = (E_1, E_2, \dots, E_q)$ is such that each edge and each vertex of G is covered by at least one of the E_i . Thus the dual hypergraph $H = (X_1, X_2, \dots, X_n)$ of $\mathcal{C}$ satisfies $L(H) = [\mathcal{C}]_2 = G$, and H is of order q .

2. We shall show that to each hypergraph $H = (X_1, \dots, X_n)$ of order q for which $G = L(H)$, we may associate a q -colouring $(S_1, S_2, \dots, S_q)$ of the vertices of $\bar{G}$. Indeed, denote by E^k the set of vertices of H which belong to exactly k of the sets X_i . We have

$$q = |E^1| + |E^2| + |E^3| + \dots$$

To each $e \in E^1$, which belongs to exactly one set $X_{i(e)}$, associate the 1-clique $\{x_{i(e)}\}$; to each $e \in E^2$, which belongs to exactly two sets $X_{i(e)}$ and $X_{j(e)}$, associate the 2-clique $\{x_{i(e)}, x_{j(e)}\}$ of G ; to each e of E^3 belonging to exactly three sets $X_{i(e)}, X_{j(e)}, X_{k(e)}$ associate the 3-clique $\{x_{i(e)}, x_{j(e)}, x_{k(e)}\}$ of the graph G ; etc.

We have thus defined in G a family $(E_1, E_2, \dots, E_q)$ of q cliques and it is evident that each edge $[x_i, x_j]$ of G belong to at least one of these (since $X_i \cap X_j$ contains a point of H). Denote by $\bar{S}_1$ the set of edges of G contained in the clique E_1 , by $\bar{S}_2$ the set of edges of G contained in E_2 which are not already contained in E_1 ; etc. The family $(\bar{S}_1, \bar{S}_2, \dots, \bar{S}_q)$ is then a q -colouring the vertices of $\bar{G}$.

It follows from points 1 and 2 that the chromatic number of $\bar{G}$ is equal to the least order of a hypergraph H such that $G = L(H)$.

Theorem 17. *Let G be a simple graph without isolated vertices, without triangles, with m edges; the minimum order of the hypergraphs for which G is the representative graph is $\Omega(G) = m$.*

Indeed, the graph $\bar{G}$ defined in the lemma is the clique K_m ; the minimum order $\Omega(G)$ is thus m , the chromatic number of $\bar{G}$.

Theorem 18 (Erdős, Goodman, Pósa [1966]). *Let G be a graph of order n without isolated points; then*

$$(1) \quad \Omega(G) \leq \lfloor n^2/4 \rfloor.$$

Further, for each n , this bound is the best possible.

Proof.

1. Indeed, we know (cf. *Graphs*, Theorem 5, Chapter 1) that we can always cover the edges and the vertices of a graph G by a family of 2-cliques and 3-cliques $\mathcal{C} = (E_1, E_2, \dots, E_k)$ with $k \leq \lfloor \frac{n^2}{4} \rfloor$; since $G = [\mathcal{C}]_2$ is the representative graph of the dual of the hypergraph $\mathcal{C}$, and since this dual is of order k , we have

$$\Omega(G) \leq k \leq \lfloor \frac{n^2}{4} \rfloor,$$

which gives us (1).

2. We show that for every n , we can have equality in (1).

If $n = 2k$ is even, take for G the complete bipartite graph $K_{k,k}$; since it has no triangles or isolated vertices we have, from Theorem 17

$$\Omega(K_{k,k}) = k^2 = \frac{n^2}{4} = \lfloor \frac{n^2}{4} \rfloor.$$

If $n = 2k+1$ is odd, take for the G bipartite complete graph $K_{k,k+1}$ which gives

$$\begin{aligned} \Omega(K_{k,k+1}) &= k(k+1) = \frac{(n-1)}{2} \frac{(n+1)}{2} = \frac{n^2-1}{4} \\ &= \lfloor \frac{n^2}{4} \rfloor. \end{aligned}$$

Thus we can have equality in (1).